

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
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Design HSE Case (ALARP Demonstration) Report

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Revision History

Rev No.	Date	Description of Revision
R01	08.08.22	Issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-THS-RPT-006

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

Abbreviation	Meaning
ALARP	As Low as Reasonably Practicable
FEED	Front End Engineering Design
HAZID	Hazard Identification
HAZOP	Hazard and Operability Studies
HEMP	Hazard and Effect Management Process
MAH	Major Accident Hazard
RoW	Right of Way
UKOOA	UNITED KINGDOM OFFSHORE OPERATORS ASSOCIATION

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

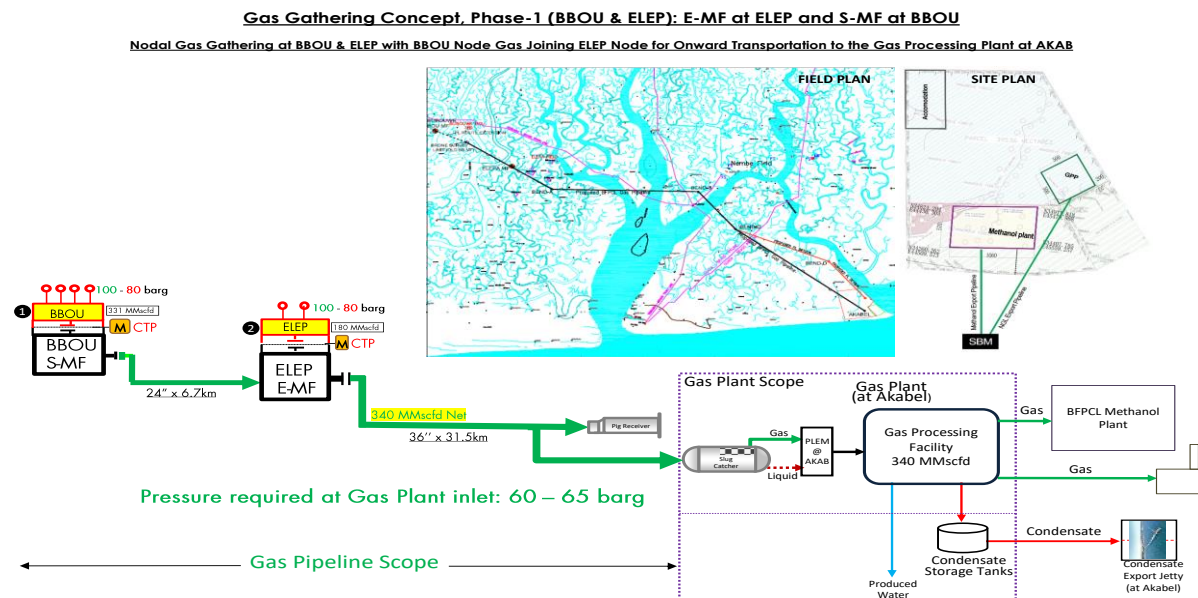


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The purpose of this Design HSE Case (ALARP demonstration) report is to demonstrate that risk reduction measures have been identified and implemented to manage the risk of Major Accident Hazards (MAH) to as low as reasonably practicable, at the FEED phase of the project.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
SR.13.14005	Shell ALARP Guide

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Hazard and Effect Management Process (HEMP)

A number of HEMP studies were carried out on the project. The findings / recommendations from these studies are summarised in this report. Tracking of the actions to close out the recommendations is done using the Project tracking system

The following HEMP activities were carried out

- a. Hazard Identification (HAZID)
- b. Hazard and Operability Studies
- c. Fire and Explosion Risk Assessment
- d. Hazardous Area Classification studies
- e. Safety Integrity Level (SIL) Classification study
- f. Bowtie analysis

3.1. HAZID Action Close Out

The HAZID was facilitated by a Technical Safety Professional and attended by relevant personnel. Current status of actions is contained in the HSE Action Tracking Register

3.2. HAZOP Study and HAZOP Action Close Out

The HAZOP was facilitated by an approved independent chairman and attended by relevant personnel. Current status of actions is contained in the HSE Action Tracking Register

3.3. Fire and Explosion Risk Assessment Action Close Out

The Fire and Explosion Risk Assessment was a desktop study and the outcome was used in layout optimisation of the project and developing the fire safety design of the facilities

3.4. Hazardous Area Classification Action Close Out

The Hazardous Area Classification study was a desktop study and the outcome was used for selection of electrical apparatus for the different hazardous classification zones in the facilities

3.5. Safety Integrity Level Classification Action Close Out

The Safety Integrity Level Classification was a desktop review, facilitated by an approved independent SIL Classification practitioner. Current status of actions is contained in the HSE Action Tracking Register

3.6. Bowtie Analysis Action Close Outs

The Bowtie Analysis was a desktop analysis. Current status of actions is part of the remedial actions in this Design HSE Case

3.7. Security, Logistics, Construction and other Risks

Security, Logistics, Construction and other hazards were identified in the HAZID and entered into the Hazards and Effects Register of the project.

4. Major Accident Hazard Management

4.1. Major Accident Hazards Identification

Major Accident Hazards (Severity 5 or Red Risk Hazards) were identified as part of the HAZID exercise. Five (5) Major Accident Hazards were identified

- Hydrocarbon Unrefined
- Logistics Hazards
- Construction Hazards
- Security Hazards
- Natural Environment Hazards (flooding, etc.)

4.2. Major Accident Hazard Risk Assessment

4.2.1. Hazard Assessment HEMP Studies

Hydrocarbon Unrefined Major Accident Hazard was assessed in various HEMP studies as noted in Section 2 of this report and resulting findings were incorporated into the design of the risk management measures adopted on the project to reduce risks to ALARP as described in Section 4.

4.2.2. Bow Ties

The Hydrocarbon unrefined Major Accident Hazard was subjected to bowtie analysis, identifying the control barriers and recovery preparedness measures including escalation / escalation factor controls. The bowtie diagram is shown in Appendix 2 and the action listing from the bowtie analysis is shown as Appendix 1

Safety Critical Elements (SCEs) & Procedures

SCEs and procedures were identified from the MAH bowties. The Hazard Summary including the listing of the SCEs and safety critical procedures are shown in Table 4-1 below.

Table 4-1: Hazard Summary of Hydrocarbon Unrefined

Hazard name:	Hydrocarbon Gas/Condensate
Top event:	Loss of Primary Containment
Threat	
	Barrier
Overpressure as result of instrument failure or human error	
	Overpressure Protection System upstream from SPDC
	Pressure Trip/blowdown system
	Pressure Safety Valve system for fire case
Mechanical impact as a result of surface lifting or excavation	
	Lifting Procedure
	Right of Way for buried pipeline
External corrosion	
	Painting for surface sections
	Cathodic protection for buried sections
Internal corrosion	
	Corrosion inhibition injection
	Corrosion inspection and Monitoring
Buoyancy	
	Design specifications
Upheaval Buckling	
	Design specifications
Sabotage	
	Community Management Procedure
	Security Management Procedure
Consequence	
	Barrier
Injuries/fatalities/asset damage from ignited release	
	Detection system (F&G at the manifold; Surveillance at Right of Way)
	Ignition Control (Hazardous area classification for the manifold)
	Fire Protection System at the manifold (Layout to prevent escalation)
	Emergency Response Procedure
Environmental pollution from unignited release	
	Detection system (Gas detection system at the manifold; Surveillance at Right of Way)
	Emergency Response Procedure

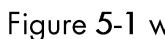
5. ALARP Demonstration

5.1. ALARP SUMMARY

The following summarises the ALARP demonstration of the project and meets the SPDC guidance on ALARP demonstration.

- a. The hazards are well understood and there is nothing novel or unusual about the scope of the project.
- b. The ALARP decision context for the scope is considered to be mostly Type A where the ALARP determination is based on applying industry standards and Shell standards, relevant guidance and good practices, supported by engineering judgment where appropriate as stated in Section 4.2
- c. Suitable risk reduction measures have been implemented.
- d. HEMP has been applied as follows:
 - Hazards have been identified and documented in the Hazards and Effects Register.
 - Process safety reviews (HAZOPs) and other reviews/workshops have been undertaken and action closed out as far as possible. Open actions are to be addressed and ensured they are closed out in the appropriate phases of the Project before handover and operation of the new installations.
 - Relevant codes and standards have been followed (Section 5.3)
- a. HEMP has been rigorously undertaken to identify, assess and manage the HSSE risks arising from the project. (Section 3 and 4). On satisfactory completion of outstanding actions, monitoring and update of the risk register, the risks will be reduced to ALARP and shall continue to be monitored to maintain ALARP.

5.2. ALARP Decision Approach

In line with Shell ALARP Guide - SR.13.14005 [1], the UKOOA decision framework shown below in  Figure 5-1 was used to determine the decision context.

The project is a pipeline project with well understood risks and is designed mostly to Shell and industry standards. Therefore, most of the HSSE related decisions and largest percentage of credible scenarios for the project fall into the "A- Routine Scenarios" category, indicating that most of the safety related decisions should be made based on accepted codes and standards, good practice and engineering judgement based on experience and knowledge. This is indicated by the position of the red band on the model above. Due to the current stakeholder management and security challenges in Nigeria, the project recognises the threat posed by this operating environment and these are being addressed separately

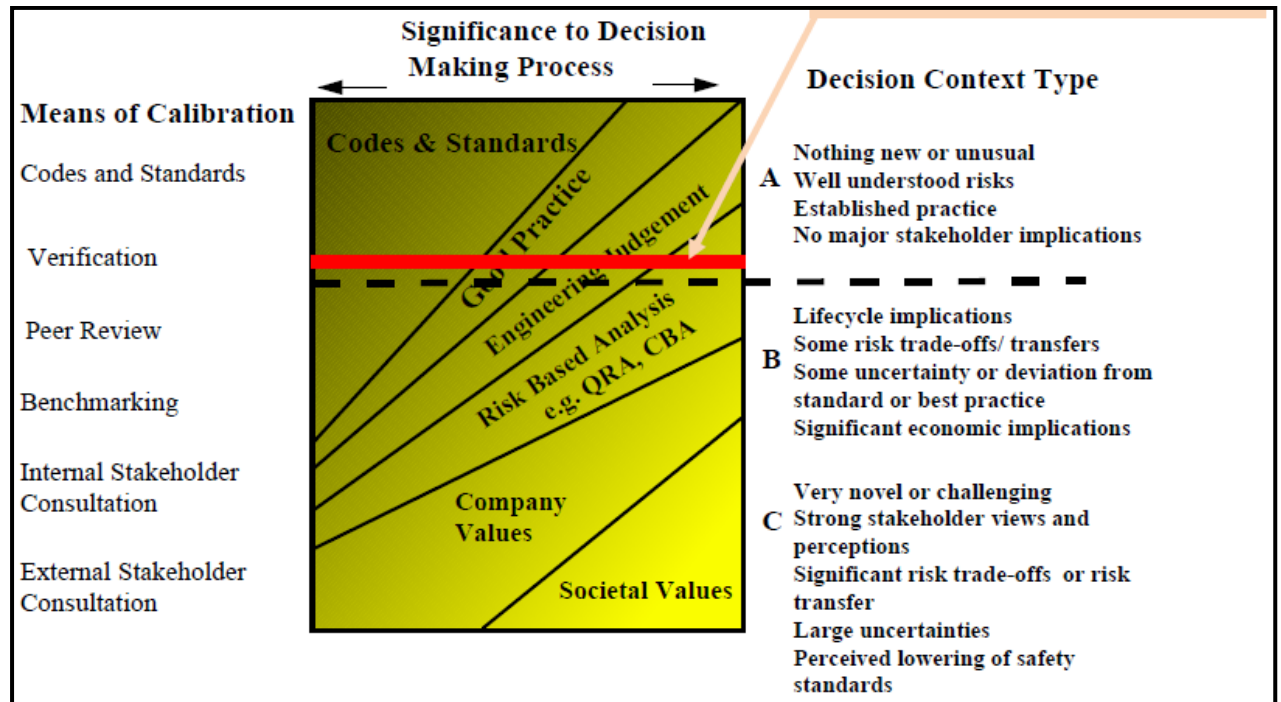


Figure 5-1: UKOOA Decision Framework

5.3. Legislation, Codes, Standards and Good Practice

A key part to achieving ALARP for many design aspects is to follow codes and standards, good practice and engineering judgment. Governing specifications, codes and standards for the design and construction has been produced in the Basis for Design for the project.

The Nigerian Regulations and associated Guidelines are identified as in other Projects and have been complied with.

5.4. Remedial Actions

Following the review of the bowtie and the assessment of what is reasonably practicable to further reduce the risk, the following remedial actions were derived to be implemented. The assignment of the responsible parties and timelines for close out of the actions will be done in the project action tracking system

- Confirm the overpressure protection system on SPDC gathering manifold and use the outcome to revalidate the SIL classification
- Develop lifting procedure to manage the risk of dropped objects resulting into loss of primary containment
- Provide Right of Way (RoW) for buried section of the pipeline to manage the risk of mechanical impact on the pipelines
- Provide painting for surface sections of the pipeline to mitigate the risk of external corrosion

- e. Provide cathodic protection for buried section of the pipeline, to manage the risk of external corrosion for the buried sections of the pipelines
- f. Provide corrosion inspection and monitoring procedure, for monitoring internal wall losses on the pipelines
- g. Design the pipeline to address pipeline buoyancy risk
- h. Design the pipeline to address upheaval buckling risk
- i. Develop a community management procedure, to improve detection of leaks on the pipelines RoW
- j. Develop a security management procedure, to mitigate the risk of sabotage
- k. Provide fire detection at the manifolds, to initiate shutdown and blowdown on confirmed fire
- l. Develop surveillance procedure for the Right of Ways
- m. Develop emergency response procedure for the pipelines

Appendix 1: Action Listing from Bowtie Analysis

Actions	Descriptions
1	Confirm the overpressure protection system on SPDC gathering manifold
2	Carry out SIL Classification of Pressure shutdown system
3	Develop a Lifting Procedure
4	Provide Right of Way for buried section of the pipeline
5	Provide Painting for surface sections of the pipeline
6	Provide Cathodic protection for buried section of the pipeline
7	Provide corrosion inspection and monitoring procedure
8	Design the pipeline to address buoyancy risk
9	Design the pipeline to address upheaval buckling risk
10	Develop a community management procedure
11	Develop a security management procedure
12	Provide fire detection at the manifolds
13	Develop surveillance procedure for the Right of Ways
14	Carryout hazardous area classification for the entire facility
15	Develop emergency response procedure

Appendix 2: Bowtie Diagram for Hydrocarbon Unrefined

